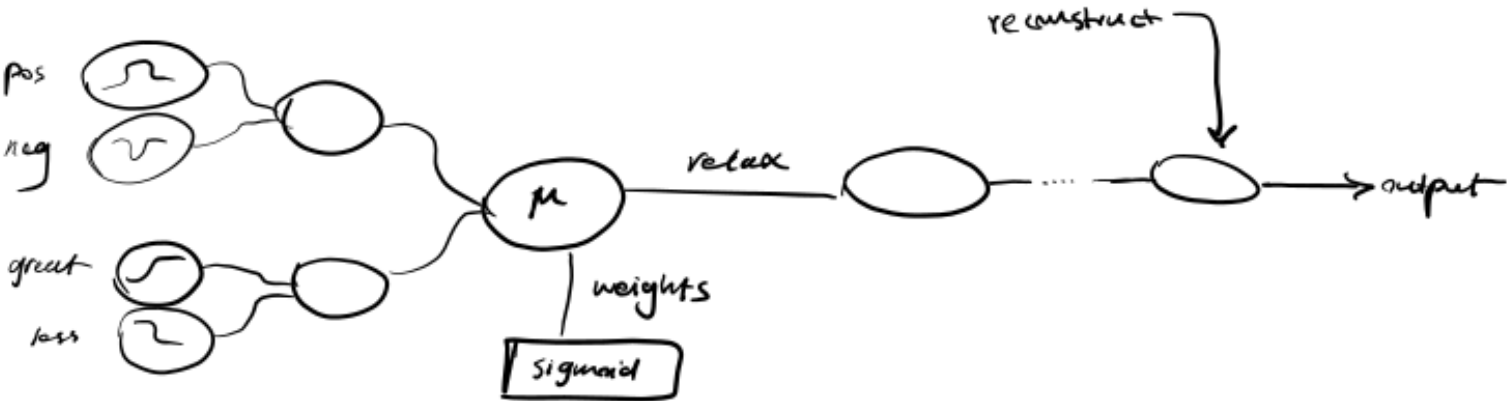
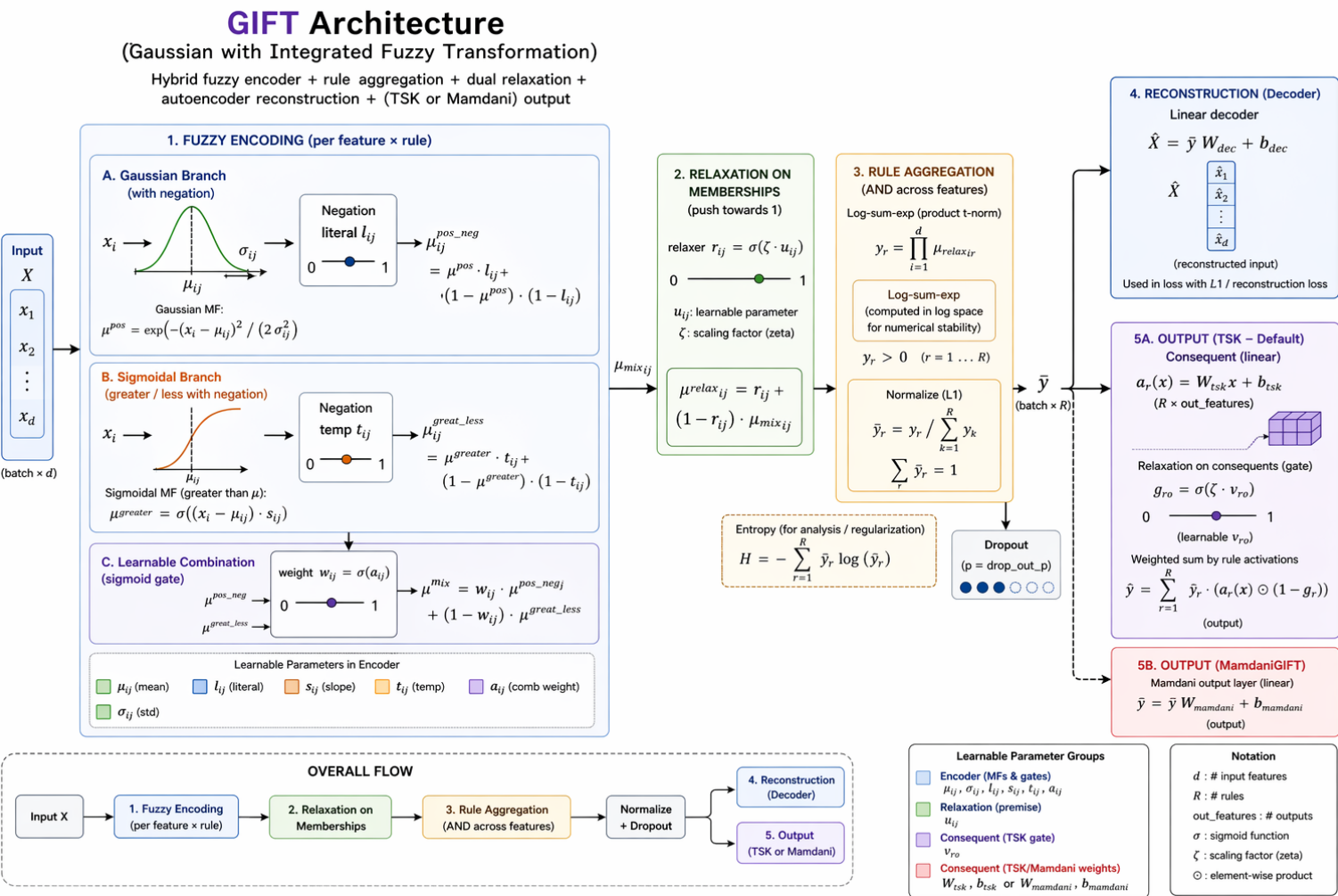


Report

changes in short:

- add the following datasets: Breast Cancer Wisconsin, Adult Census Income, Bank Marketing, Pima diabetes, Car Evaluation
- changed the structure of GIFT
- tried mamadani to compare with tsk

the current structrure as below



1. learn between pos vs neg => parameter: literal (parallel)

$$\mu_{pos_neg} = (\mu_{pos} * literal) + (1 - \mu_{pos}) * (1 - literal)$$

2. learn between greater vs less => parameter: temp (parallel)

$$\mu_{great_less} = (\mu_{greater} * temp) + (1 - \mu_{greater}) * (1 - temp)$$

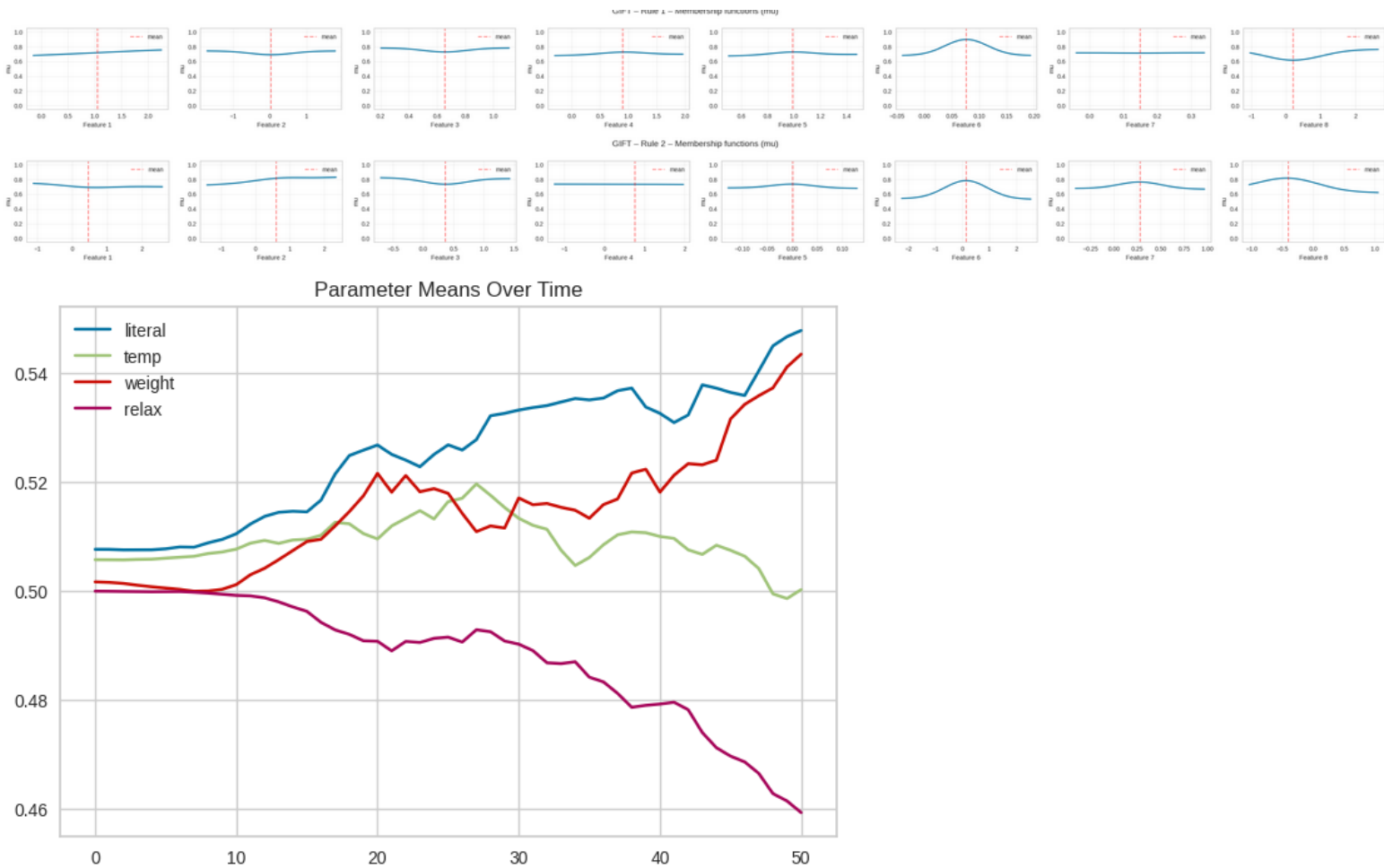
3. then combine the two mu's => parameter: weight: output of sigmoid

$$\mu = weight * \mu_{pos_neg} + (1 - weight) * \mu_{great_less}$$

4. and then relaxation => parameter: relaxer

$$\mu = relaxer + (1 - relaxer) * \mu$$

*** membership function plot and the evaluation of mentioned parameters (pima dataset with k=2):



Results (tsk & mamdani) and compare to litanfis

k = 2 & 5 runs

Breast Cancer

litanfis(tsk) => 97.71

gift (tsk) => 97.35

gift (mamdani) => 98.07

Adult Income

litanfis (tsk) => 85.89

gift (tsk) => 85.38

gift (mamdani) => 85.98

Bank Marketing

litanfis (tsk) => 90.05

gift (tsk) => 90.34

gift (mamdani) => 89.97

Pima Diabetes

litanfis (tsk) => 76.88

gift (tsk) => 76.75

gift (mamdani) => 75.84

Car Evaluation

litanfis (tsk) => 92.61

gift (tsk) => 92.9

gift (mamdani) => 88.07

Note: Parameters initialization:

literal: Initialized with small random numbers drawn from a standard normal distribution and scaled by 0.1, resulting in values near zero. This makes the sigmoid of literal start around 0.5, giving equal weight to the membership and its negation initially.

temp: Same as literal: small random values ($\text{normal} \times 0.1$), so its sigmoid is about 0.5 initially, balancing the “greater than mu” branch and its negation.

comb_weight: Also small random values ($\text{normal} \times 0.1$), causing the sigmoid of comb_weight to be near 0.5. Thus the two membership branches (Gaussian-negation and sigmoidal-negation) are equally combined at the start.

relax: Initialized to all zeros. The sigmoid of $\text{relax} * \text{zeta}$ is therefore 0.5, leading to the relaxation formula $\mu = 0.5 + 0.5 * \mu$ (a mild push of the combined membership toward 0.5).